

# FileSort Pro

AI-Powered File Organization System Idea Brief

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## The Problem

Every computer user faces the same problem: files downloaded from the internet end up scattered randomly across folders. CVs mixed with images, PDFs mixed with installers, important documents buried under random screenshots. Users waste hours searching for files and manually organizing their Downloads folder only for it to become messy again within days. There is no simple, automatic solution that works across both the browser and the operating system without requiring technical knowledge.

## The Solution FileSort Pro

FileSort Pro is a two-part automation system that works silently in the background and organizes every file the moment it is downloaded without any user input.

|                                        |                                     |                            |                                          |
|----------------------------------------|-------------------------------------|----------------------------|------------------------------------------|
| <b>800+</b><br>Files sorted in testing | <b>13</b><br>Sorting rules built-in | <b>2 min</b><br>Setup time | <b>Zero</b><br>Manual work after install |
|----------------------------------------|-------------------------------------|----------------------------|------------------------------------------|

## How It Works

### Part 1 — Chrome Browser Extension

- Monitors every file downloaded through Chrome or Edge in real time
- Matches filename and extension against 13 built-in sorting rules
- Shows an instant desktop notification: file name and suggested folder
- Dashboard popup shows activity history and lets user toggle rules on/off

### Part 2 — Windows Background App (System Tray)

- Runs silently in the background, starts automatically on every Windows login
- Watches the Downloads folder using a real-time file system watcher
- Physically moves files to the correct organized folder automatically
- Logs every action with timestamps — full audit trail
- System tray icon shows files sorted count, pause/resume, open log

## Technology Stack

|                          |                                                                           |
|--------------------------|---------------------------------------------------------------------------|
| JavaScript / Chrome APIs | Browser extension, service worker, chrome.downloads, chrome.notifications |
| Python 3                 | System tray app, file watcher, file mover, config manager                 |
| Watchdog library         | Real-time file system event detection on Windows                          |
| Pystray + Pillow         | System tray icon rendering and menu                                       |
| PowerShell               | Windows toast notifications, startup registry, installer                  |
| Regex pattern matching   | Intelligent file name analysis for CV, LOR, SOP, invoice etc.             |

## Future: AI/ML Upgrade Opportunities

The current version uses rule-based pattern matching. The next version can be significantly more powerful with AI:

- Train a lightweight classifier (Random Forest or fine-tuned BERT) on filename patterns to auto-learn new categories
  - NLP-based content analysis — read PDF/DOCX content to determine category, not just filename
  - Personalization engine — learns each user's own folder preferences over time
  - Anomaly detection — flags suspicious files (malware patterns, unusual extensions)
  - Cross-platform expansion — macOS, Linux using the same Python core
  - Cloud sync integration — auto-upload sorted files to Google Drive / OneDrive by category
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## Market Opportunity

- 3+ billion Chrome users worldwide — Chrome Web Store distribution
  - 1.4 billion Windows PCs — Microsoft Store or direct download
  - Target users: students, professionals, remote workers, freelancers
  - Monetization: free basic tier, Pro tier (\$3/month) with custom rules + cloud sync
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